



SECOND SERVE

AI-Driven Circular Food Redistribution Platform

A citizen-centric AI system for
urban well-being.

The Challenge



Core Conflict: Urban cities generate massive daily surplus, yet vulnerable populations, animal shelters, and farmers face acute resource shortages.

Systemic Failures:

- **Lack of Structure:** Restaurants lack safe systems to redistribute excess.
- **Invisibility:** NGOs lack real-time visibility of available surplus.
- **Inefficiency:** Manual coordination and poor safety monitoring lead to avoidable waste.

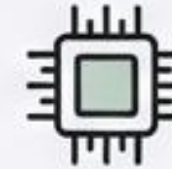
Motivation



Environmental: Reduce landfill methane emissions and strengthen circular economy practices.



Social: Address urban hunger and support animal shelters with safe surplus.



Technological: Shift from manual coordination to data-driven, AI-enabled redistribution.

Long-Term Vision

Statement: To build an AI-enabled ecosystem that enhances national well-being, social stability, and resource equity.

Social Stability • Environmental Sustainability
• **Responsible Urban Resource Management**

Module A: Proposed Method

The Intelligence Layer

ML Freshness Prediction

- Trained on spoilage data, weather, storage.
- Output: ● 'Safe/ ● Risk' classification & edible time.

Intelligent Matching

- Algorithms prioritize high-need zones.
- Minimizes delivery distance/time.

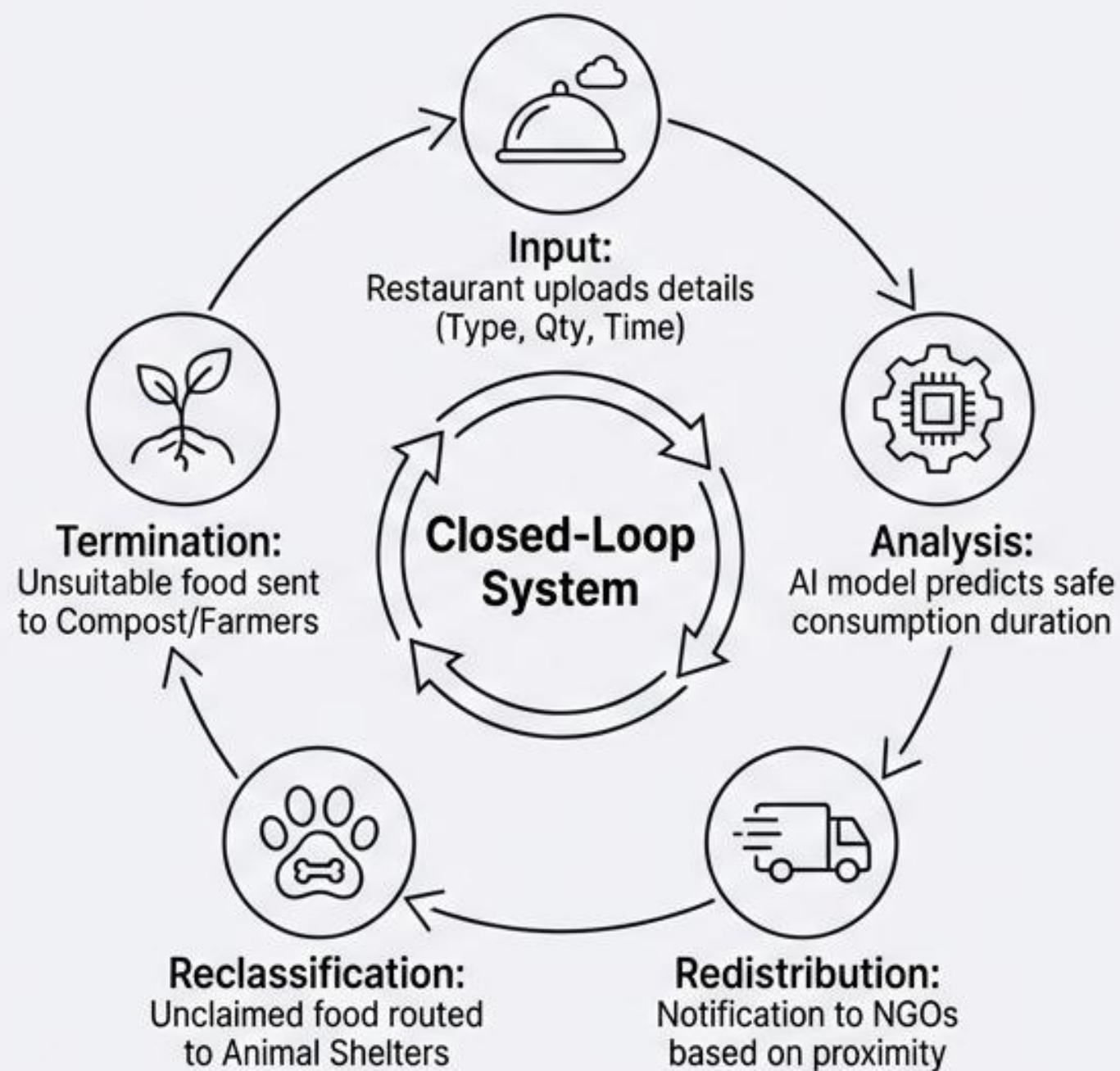
GenAI Layer

- Automated notifications & impact dashboards.
- NGO coordination chatbot.

Dynamic Reclassification

- ML transitions categories: Human → Animal → Compost.

The Real-World Workflow



Module B: Ecosystem Scope

Deployment

Target Users

- Restaurants/Cafes
- NGOs/Food Banks
- Low-income Individuals
- Animal Shelters
- Farmers



Environments

- Urban Smart Cities (Mumbai Pilot)
- Corporate Food Courts
- Event Venues
- College Campuses



PROJECT FOUNDATION & FUTURE

Datasets

Sources:

- Historical shelf-life datasets
- FSSAI food safety guidelines
- Restaurant surplus logs & Municipal waste data

Dynamic Inputs:

- Weather API & NGO geolocation

Formats:

- Structured tabular, Geospatial, Time-series



Experiments

Technical Verification:

- Accuracy testing of freshness models
- Classification performance & Debugging

Impact Verification:

- Waste reduction % measurement
- Time-to-redistribution analysis

User Verification:

- NGO feedback surveys
- Pilot testing: Manual vs. AI matching



Key Innovations

Novelty

Core Differentiation:

- AI-powered shelf-life prediction + Circular logic

The Triple-Win:

- Multi-tier utilization: Human -> Animal -> Compost

Systemic Advance:

- Social welfare integration with AI decision intelligence
- Adaptive learning system



Roadmap

Scalability Potential

Geographic:

- Tier 2/3 cities & Municipal partnerships

Sectoral:

- CSR, Weddings, Supermarkets

Technical:

- API integration with delivery platforms
- National sustainability dashboards

Crisis Response:

- Disaster-relief redistribution

